

Dhruvil Shah

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EDUCATION

• Dwarkadas J. Sanghvi College Of Engineering (DJSCE) | Mumbai, India

Nov'2021 - May'2025

- ❖ Bachelors of Technology (Artificial Intelligence & Machine Learning)
- ❖ CGPA - 8.81

RESEARCH TRAINING

• Vizvara AI Labs | [RL in Production Bootcamp](#)

Jun'26–Current

- ❖ Pursuing a six-month program covering MDPs, DP/MC/TD learning, DQN, policy gradients, actor–critic, TRPO/PPO, RLHF, reward modelling, KL regularisation, GRPO/RLVR, preference optimisation and agentic RL.
- ❖ Coursework includes reward-hacking analysis, verifier-based training, RL environment design and comparative post-training methods including DPO, KTO, ORPO and SimPO.
- ❖ Currently studying and implementing IRIS-style world models using VQ-VAE representations, autoregressive Transformer dynamics and actor–critic learning in imagined trajectories, including the Catch and VizDoom tracks.

WORK EXPERIENCE

• Rezinix | Data Engineer

Feb'2026 - Current

- ❖ Working in a lead-like cross-functional role across projects, contributing to system design, technical execution, and client-facing discussions.
- ❖ Designed and led development of a self-hostable custom form OCR platform.
- ❖ Integrated an AI response pipeline for [FleekAI](#), an AI automated marketing platform.
- ❖ Contributed to the 3D model generation pipeline for [JewelTech](#).

INTERNSHIP EXPERIENCE

• NotSoHuman.ai | Machine learning intern

Cloud Fine-Tuning, ComfyUI, FLUX Dev & Schnell

- ❖ Tweaked hyperparameters and explored fine-tuning methods for FLUX models to enhance image generation quality
- ❖ Experimented with various generation techniques to generate realistic images of individuals in imaginative settings

INDEPENDENT AI SAFETY EXPERIMENTS & WRITING

• Reasoning Effort Under Indirect Prompt Injection [🔗](#) | Independent Pilot Study

Jul'26

- ❖ Built a pinned gpt-oss-20b Inspect/AgentDojo evaluation across banking, Slack, travel and workspace environments, producing 9,348 finalised scenario-level rows.
- ❖ Compared low, medium and high reasoning effort under clean tasks, fixed injections and attack-template search; found no statistically resolved safety effect in the completed conditions.
- ❖ Stopped confirmatory transfer and defence stages when the hosted notebook's 12-hour runtime cap made the remaining run infeasible and restarting would discard in-memory state; published the work explicitly as an incomplete pilot.

• KL Regularisation Under Reward Misspecification [🔗](#) | Independent Experiment

Jul'26

- ❖ Designed and ran a controlled tabular study spanning 6,144 stationary and time-aware policies, 24 KL strengths, 64 replicas and 36.86M training episodes.
- ❖ Diagnosed a policy-class confound in the original design, then introduced time-aware state and an exact finite-horizon dynamic-programming oracle before interpreting results.
- ❖ Found that no tested KL strength eliminated proxy excess; reported the result as model-organism evidence rather than a universal claim about frontier RLHF.

• From Benchmarks to Barriers [🔗](#) | Independent AI Strategy Article

Jul'26

- ❖ Synthesised evidence from both safety studies into a framework for deciding when evaluation results should change deployment conditions rather than remain benchmark scores.
- ❖ Connected adaptive evaluations to defence-in-depth measures including least-privilege tool access, independent monitoring, human approval and staged deployment authorisation.
- ❖ Separated observed evidence from strategic inference, documented the reasoning study's incomplete confirmatory stages and identified the follow-up evidence required for stronger decisions.

PROJECTS

• Secure Data Transmission | Final year project

Feb'25 | *Embedded Python, Rust, sklearn, Flask, Docker, OpenCV, JS*

Developed an innovative file encoding system that converts files into encoded video for secure cross-platform sharing. This project leverages **ML-based pixel clustering** for region-based video merging, ensuring data integrity while creating natural-looking merged videos. **Rust** is employed for high-performance processing to etch the source file into a video, complemented by **Python and Computer Vision** for sophisticated region-based merging techniques to obscure the etched data within everyday video content. **Docker** containerization is implemented to ensure reproducible development and consistent deployment across platforms. This project won my team a college level event named Tech expo event.

- ❖ First author and lead contributor — conceived the core idea and led implementation and ideation efforts.
- ❖ Winner of the college-level Tech Expo event for innovation and technical excellence in cybersecurity domain.

• EmotiSense | Full Stack Web Application ([Publication](#))

Dec'23 | *Embedded Python, Tensorflow, Flask, Librosa, OpenCV, JS*

Developed an end-to-end website to track real time visual emotions using **CNN** model and audio based emotions using **RNN** model. The models were trained on open source datasets. Provides **video mapped audio** and **video timelines** as well as **video mapped audio frequency** and **intensity graph** based on emotions predicted by the models. Based on the session data overall behavior is concluded. This project has been **published** in the "Library Progress International" Journal.

• FLAN-T5 StatGuide | Full Stack Web Application

Nov'23 | *Embedded PyTorch, Flask, Google FLAN-T5, Transformers, React*

A systematic user centered platform for solving statistical hypothesis questions based on t-test, z-test, f-test using **Google's FLAN-T5 model fine-tuned** using **transformers** library on **custom dataset**. Provides accurate answers compared to other LLMs by combining novel prompt engineering technique and LLM predictions with statistical hypothesis algorithms to deal with the limitations of **vector embeddings**.

- SKILLS
- **Programming Languages :** *Python, C, C++, Rust, JavaScript, TypeScript*
 - **Frameworks & Libraries :** *Flask, Django, React, Next.js, Three.js, Node.js, Express.js*
 - **AI/ML Tools & Libraries :** *TensorFlow, PyTorch, Scikit-learn, OpenCV, Librosa, Transformers, NLTK, Keras*
 - **Data Analysis & Visualization :** *Pandas, NumPy, Matplotlib, Seaborn, Microsoft Excel*
 - **Web Development :** *HTML, CSS, Bootstrap, TailwindCSS*
 - **Databases :** *MySQL, SQLite, PostgreSQL, MongoDB, Supabase, Convex*
 - **Big Data Tools :** *Hadoop (HDFS, MapReduce), Spark (RDDs, MLlib), Hive*
 - **DevOps/MLOps :** *Docker, Git, MLflow, REST APIs*
 - **Other Tools & Technologies :** *Linux, ROS (Robot Operating System), CUDA (C/C++ & Python), ComfyUI, Blender*
 - **AI Safety & Evaluation :** *Reward misspecification, proxy gaming, reward hacking, KL regularisation, indirect prompt injection, instruction-hierarchy robustness, agent/tool-use evaluation, safety-utility analysis*
 - **RL & LLM Post-training :** *MDPs, Q-learning/DQN, policy gradients, actor-critic, TRPO/PPO, RLHF, reward modelling, GRPO/RLVR, DPO, KTO, ORPO, SimPO*
 - **Agentic & Model-Based RL :** *Custom environments, verifiable rewards, world models, VQ-VAE representations, autoregressive dynamics, policies trained through imagined rollouts*
 - **Research Tooling :** *PyTorch, Transformers, Gymnasium, Inspect AI, AgentDojo, marimo, NumPy, pandas, Matplotlib, controlled ablations and exact-oracle evaluation*

Certificates

- **Python for Computer Vision with OpenCV and Deep Learning by Jose Portilla** *Udemy*
 - **Python for Machine Learning & Data Science Masterclass by Jose Portilla** *Udemy*
 - **The Ultimate React Course 2024: React, Next.js, Redux & More by Jonas Schmedtmann** *Udemy*
 - **ROS for Beginners: Basics, Motion, and OpenCV by Anis Koubaa** *Udemy*
 - **Fundamentals of Deep Learning** *Nvidia*
 - **Fundamentals of Accelerated Computing with CUDA Python** *Nvidia*
 - **Fundamentals of Accelerated Computing with CUDA C/C++** *Nvidia*